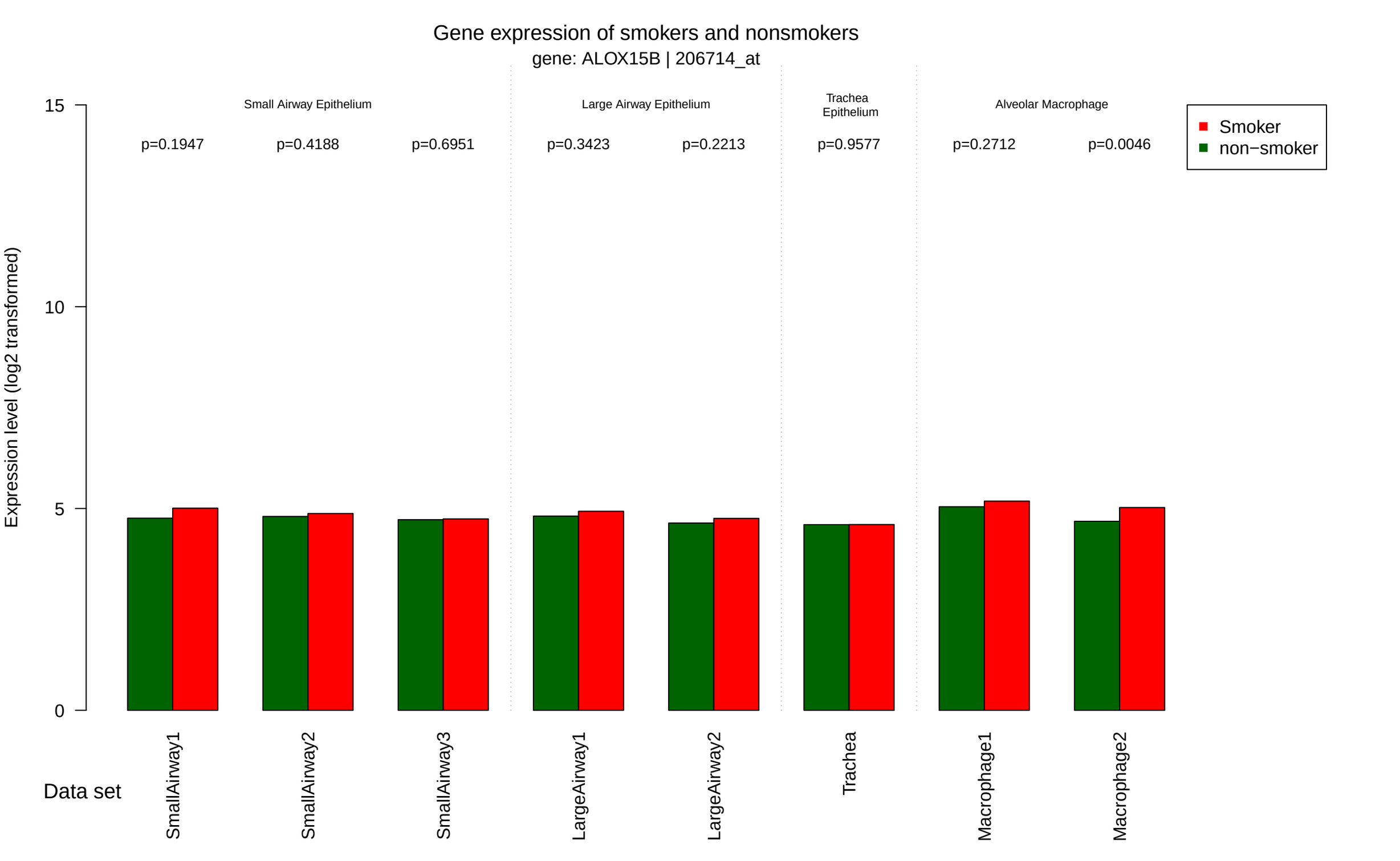
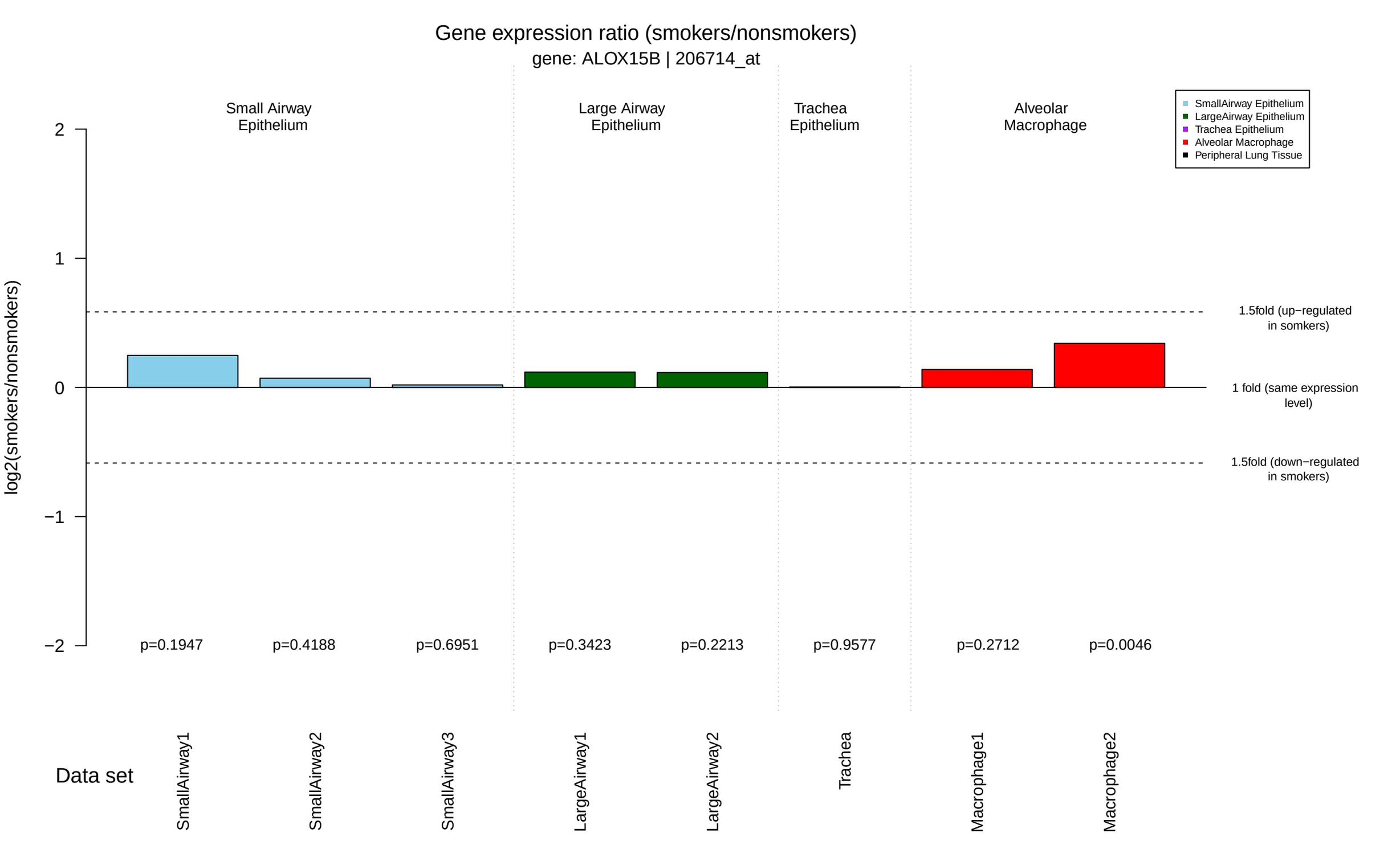
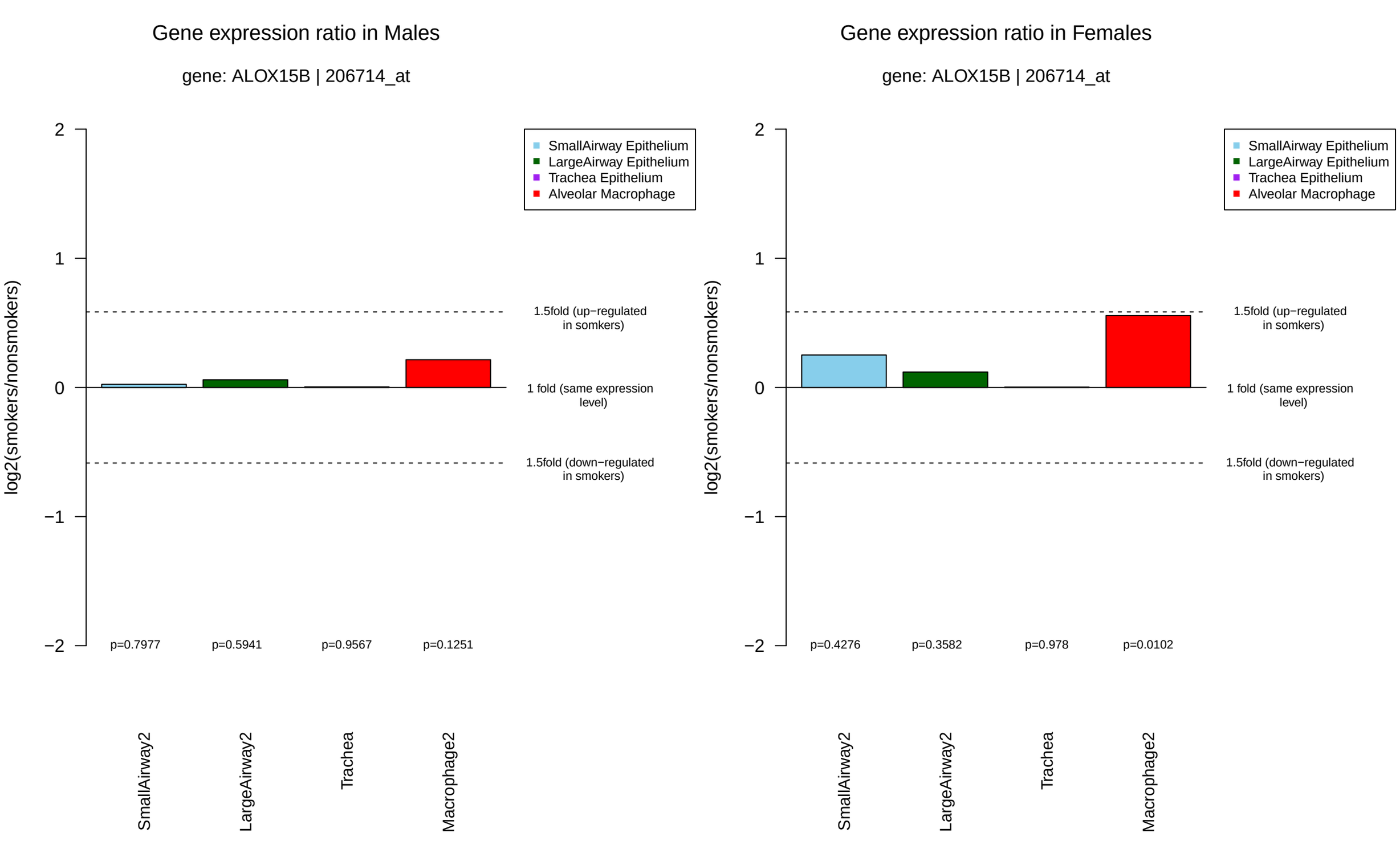
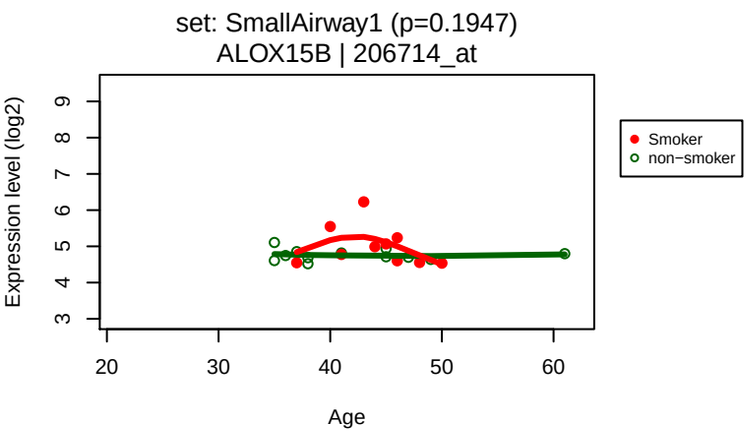
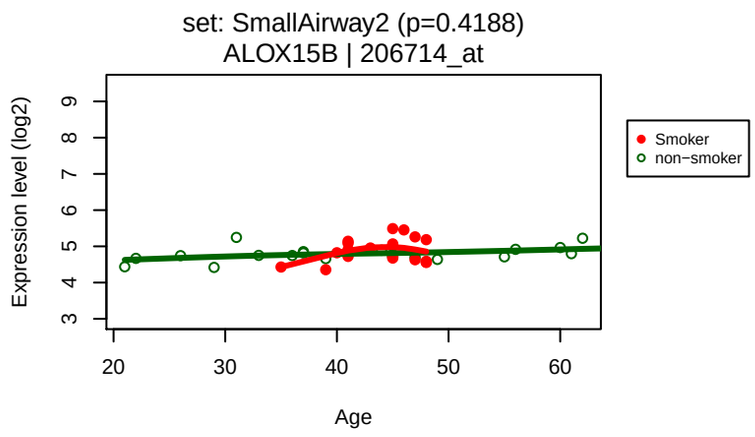
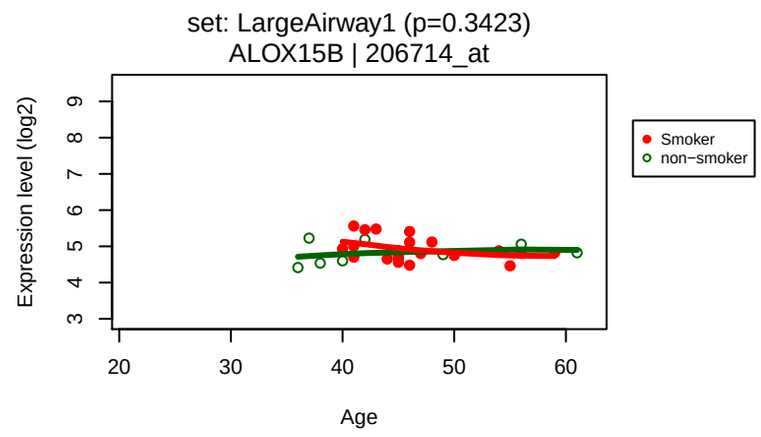

Gene expression of smokers and nonsmokers
gene: ALOX15B | 206714_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1947
p=0.4188
p=0.6951
p=0.3423
p=0.2213
p=0.9577
p=0.2712
p=0.0046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

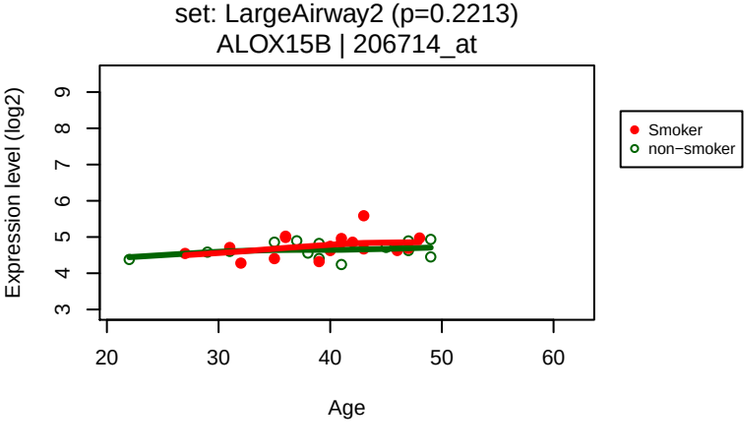

Gene expression ratio (smokers/nonsmokers)
gene: ALOX15B | 206714_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1947
p=0.4188
p=0.6951
p=0.3423
p=0.2213
p=0.9577
p=0.2712
p=0.0046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

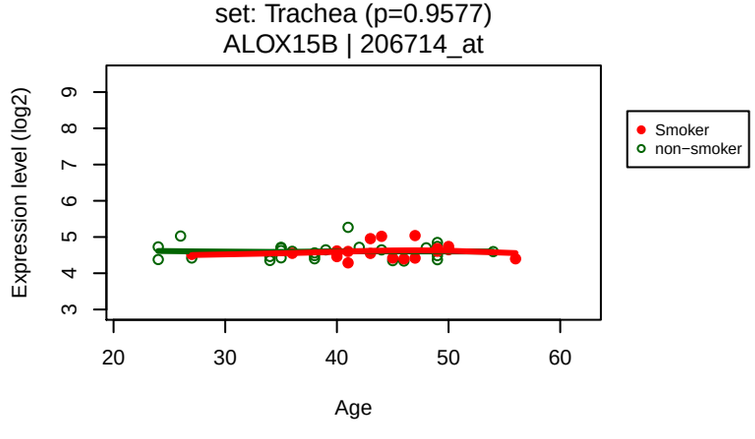

Gene expression ratio in Males
gene: ALOX15B | 206714_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7977
p=0.5941
p=0.9567
p=0.1251
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX15B | 206714_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4276
p=0.3582
p=0.978
p=0.0102
SmallAirway2
LargeAirway2
Trachea
Macrophage2

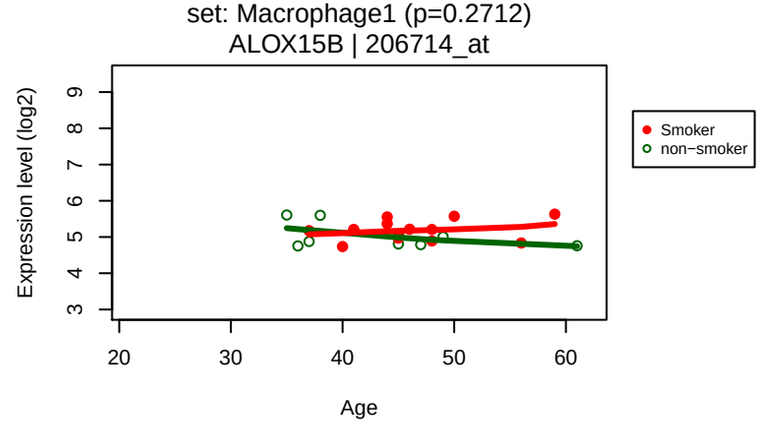

set: SmallAirway1 (p=0.1947)
ALOX15B | 206714_at
Expression level (log2)
Age
Smoker
non−smoker

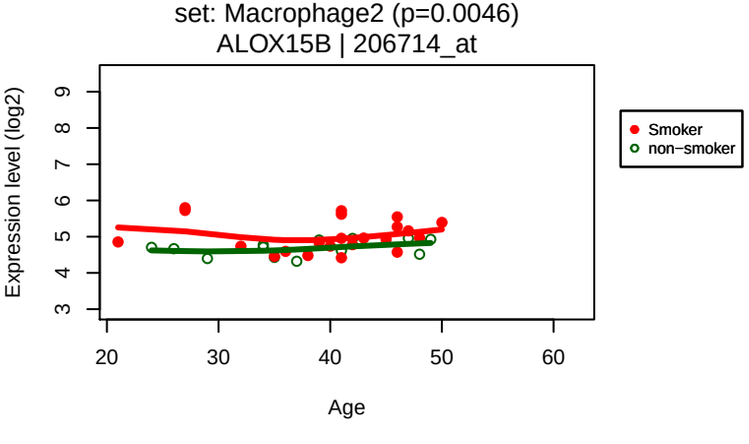

set: SmallAirway2 (p=0.4188)
ALOX15B | 206714_at
Expression level (log2)
Age
Smoker
non−smoker

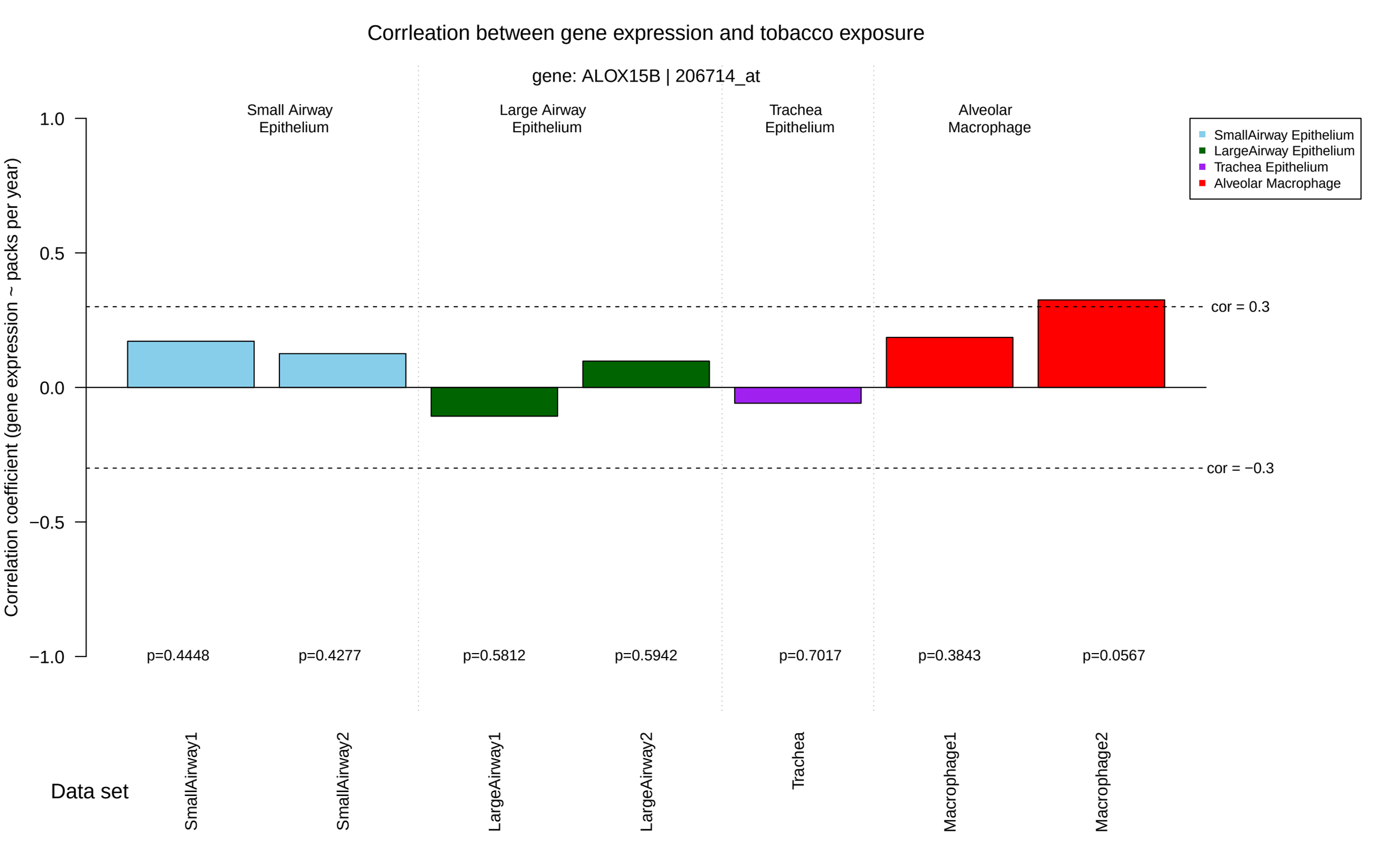

set: LargeAirway1 (p=0.3423)
ALOX15B | 206714_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2213)
ALOX15B | 206714_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9577)
ALOX15B | 206714_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2712)
ALOX15B | 206714_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0046)
ALOX15B | 206714_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX15B | 206714_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4448
p=0.4277
p=0.5812
p=0.5942
p=0.7017
p=0.3843
p=0.0567
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2